

Manual

Energy Data Collection / Energy Management EMG-ARC 8.1

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1 Overview of Energy Management

Purpose

This component forms the basis for applying HYDRA as the energy management system used for operations. DIN EN ISO 50001 defines the requirements that apply to a production company's energy management used for operations. The standard replaces DIN EN 16001 as of 2012. HYDRA as an MES system offers the basic principles and tools that allow a system of this type to be designed with the support of IT. One primary aspect concerns how energy values are monitored, measured and analyzed. However, HYDRA also includes additional functions that can be used to define measures or to prepare documentation.

You use the function package if:

- You would like to manage energy counters in HYDRA, and
- You would also like to record the energy or efficiency values for these counters. Data is collected automatically via the HYDRA-PCC (process communication controller) or manually via read-out plans (EMG-AME package).
- The package is the basis for all further additional packages in energy management.

Integration

HYDRA PCC is used to collect data for automatic collection. It is supplied separately and is not a part of the package.

To analyze the data, this basic package only includes a list of chronologically organized raw data in tabular form, which at the same time is also the correction and editing function. Other additional packages must be used for analyzing and monitoring.

In connection with the use of the WRM module, the status change and maintenances can also be collected on the counter resources and be analyzed in a history.

Features

- Master data editing and counter configuration
 - Administration of the real energy meters (counters) as resources. Generates logical, calculated counters used to combine several collection points to efficiency groups. Differentiates counted values by shifts, days or months.
 - Functions for version administration of the counters and with which to reset counters when replacing counter devices.
 - Configuration and definition of the counter channels to record consumption and to measure energy

- Administration and posting of resource states for the meters.
 - Assigns counters to workplaces or to other properties used for logistics.
 - Manages properties used for logistics such as industrial space, halls and buildings as well as the machines and aggregates as resources.
 - Hierarchical property management: Breaks down buildings and plants, machines and energy counters into a tree-like structure.
- Data entry, data editing, data keeping
 - Imports counted values via the process communication controller.
 - Stores the counted values chronologically in the database. Posts and accumulates counted values to resources.
 - Option to correct counter states and thus also the consumption data by manually recording the corrected records or editing them in the postings.

2 Workplace and Resource Configuration

Overview

HYDRA menu	Master data → Resources → Resource configuration Master data → Workplaces/machines → Workplace configuration
FEDRA menu	Detailed Scheduling → Master data → Resource configuration
Transaction code	res
Function authorization	mdres mdresgenh for fields in combination with Test Equipment Management

Available user fields		
Where?	Object type/user field key	Source (type)
Tab <i>User fields</i>	<Res.type*>/depending on data record	Resource (MF-D)
Table	RES/SYSTEM	Resource (MF-D)

*) <Res.typ> = resource type

The resource configuration is the central function to manage resources in the MES.

Purpose

This application manages the master data of workplaces/machines and other resources (tools, DNC resources, etc.). The resource type classifies resources. Each resource type is also linked to specific functions and applications, which provide further functionalities of the MES for resources of the specified type.

Integration

Use this application to view the resource information of all resource types available in the system. The resource type also specifies how and if data records can be edited. Depending on the resource type, you cannot edit all fields or create and delete all resources.

Based on the resource type, the MES also includes further applications that are especially tailored to these types. The *machine data collection* application package, for example, is based on resources of the type "machine".

In addition to the resource configuration, the *resource overview* application is available. You cannot use the *resource overview* application to edit data. This application only allows administrative operations for the daily handling of resources such as the stock transfer of resources.

Requirements

Create a year model/shift calendar prior to creating a workplace or machine. If you want to use the various resource types effectively, you also need the advanced licenses for these types.

Selection criteria

The application provides the following selection criteria:

Resource from ... to ...

This selection criterion refers to the resource. You can also use wildcards (placeholders *).

Short name

Short name of the resource. Only relevant for resources of type MNR.

Resource type

Type of resource.

Workplaces and machines always have the resource type MNR. But you can assign individual resource types to the other resources by configuration. Predefined resource types include:

DNC	NC/DNC program
DOC	Document
ENE	Energy meter
ENT	Removal device
ENT	Removal device
MNR	Workplace/Machine
PAC	Packaging, transportation container
PRM	Test and measuring equipment
PER	Production staff / general
PRU	Setup staff
TEM	Tempering equipment
VOR	Device
WNR	Tool

We recommend using the predefined resource types.



The displayed detail resource information varies with the resource selected in the table overview.

Name

Name of the resource.

Group

Workplace/machine group of the resource. Only relevant for resources of type MNR.

Cost center

Cost center of the resource.

Short name

Short name of the resource.

Resource family

Family the resource is assigned to.

Responsibility area

Responsibility area the resource is assigned to.

Storage location

Regular storage location of the resource.

MD user fields

MD user fields 1- 6 of the resource. If you select a resource family in the selection panel, the application shows the field names according to the assigned user field definition.

Field descriptions

This detail application includes four main tabs:

- Resource configuration
- Resource list
- Resource attributes
- DNC versions

Main tab *Resource configuration*

Here, you can define the configurations and master data of resources.

General tab

Resource type

Resource type of the resource. The system delivery includes some default resource types. Create additional resource types in the application .

Resource

Enter the number of the resource or workplace to be collected in this field.

The resource type also specifies the maximum number of characters that are allowed for the resource number:

- Resources of the type MNR: a maximum of 8 digits
- Resources of a type <> MNR: a maximum of 20 digits

Permitted characters: ABCDEFGHIJKLMNOPQRSTUVWXYZ1234567890_.-+#. Do not use spaces and other special characters. For technical reasons, you can enter * (asterisk) and % (percent), but they are nonetheless not permitted because they are not valid characters. When you exit the input field, the system automatically converts lower case letters into CAPITAL LETTERS.

Please note for workplaces/machines (resource type MNR):

For technical reasons, the system does not check the maximum number of digits allowed for resources of the type MNR. For this reason, make sure that the resource number length (= workplace/machine number) does not exceed 8 digits.

Please note: If you set the resource type MNR before entering the resource ID (machine number), the GUI only allows you to enter eight digits.

If you select the option "numeric machine number" (basic parameter settings) for use with DOS terminals, you must ensure that the resource number (= workplace/machine number) only includes numerical digits and that its length is exactly 8 digits. If necessary, prefix leading zeroes to the number to extend it to eight digits, when creating the workplace/machine.

Short name

Short name of the resource. Only use this field with workplaces/machines (resources of the type MNR).

Name

Use this field to assign a short, unique name to each resource. Reports and overviews as well as terminal dialogs show this name, which is also useful for orientation purposes.

Responsibility area

Use responsibility areas to restrict the data users can view in different evaluations/reports. Users can only view the data they are allowed to according to their responsibility area authorization.

The *responsibility area* field can also remain empty. In this case, the resource is always displayed regardless of the user's assigned responsibility authorizations.



If you leave the *responsibility area* field empty, the system automatically enters the value "--DEFAULT--" in the field. Resources including this value are always displayed regardless of the user's assigned responsibility authorizations.

Cost center

This field includes the cost center the resource is assigned to.

Inventory number, engraving number, drawing number, manufacturer, owner

Additional information in form of comments.

Acquisition date, acquisition costs

Additional information in form of comments.

Configure the currency for the entire system in the basic settings.

Storage location

Location where the resource is stored when it is not being used (original storage location).

In connection with the Material and Production Logistics (MPL) product group, this field specifies a material buffer. If you log on an input batch, the logged on input batch(es) will be transferred from the previous material buffer to the material buffer entered in this field (upstream of the machine).

Delivery date, start-up date, guarantee date

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

External designation, resource type designation, usage, purchase order number

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Supplier and party in charge including detail fields

Additional information in form of comments. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Workplace configuration tab

This tab is only available if you select a resource of the type "MNR".

Workplace master data**Workplace category**

N Machine

P Workplace

Defined as machine or workplace. If you exclusively use BDE and/or MDE and PDV, the two categories are identical as regards processing.

J Machining center (BDE-BEA only)

The "Machining center" category and its functionality are described in detail in the BDE-BEA product documentation.

L Line (MDE-SFL only)

A Aggregate (MDE-SFL only)

The categories "Aggregate" and "Line" and their functions are described in detail in the MDE-SFL product documentation.

Q CAQ inspection station

Workplace is defined as mere CAQ inspection station and does not affect BDE or MDE statistics.

R Coil-based manufacturing (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

S Cutting unit (only for coil-based manufacturing)

This type controls specific functions for the coil-based manufacturing.

D Parallel output batches (only MPL)

You can produce parallel output batches on the machine for an operation that requires batch management.

C Packing station (only MPL)

You can use specific posting functions of the machine to represent a packing station. The functions are described in detail in the AIP-LCS product documentation.

M Melting aggregate

This option defines a machine as melting aggregate in terms of composition.

F Laboratory/in-production inspection

This workplace is configured as inspection station. The inspection points are displayed, which are assigned to this workplace or machine group of this workplace because of the higher-level inspection point. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

```
<MachineSpecifiedLayout>True</MachineSpecifiedLayout>
```

W Goods receipt inspection

This workplace is configured as inspection station. The goods receipt inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

```
<MachineSpecifiedLayout>True</MachineSpecifiedLayout>
```

K Calibration

This workplace is configured as inspection station. The calibration inspection points are displayed, which are assigned to this workplace or machine group of this workplace. You must activate the workplace-specific layout here. Use the following parameters for activation in the AIP layout file "globaldefines.xml".

```
<MachineSpecifiedLayout>True</MachineSpecifiedLayout>
```

Workplace type

E Single workplace (SWP)

G Group workplace (GWP)



Group workplaces are workplaces without machine data collection or MDE evaluations.

In case of group workplaces, you cannot post to resource performance accounts in an operation-related manner with postings based on the current machine status. Only main production times (RPA 11) are recorded. You must define a status with the control indicator "production" in the .

The system does not generate for group workplaces. Therefore, MDE evaluations that evaluate MDE log records are not possible.

Like single workplaces, you can assign group workplaces to terminals. In this case, you have to make sure that the is set to operation mode "BDE" or the option *Processing* is set to "BDE processing" in the .

External workplace

This field identifies external workplaces. Currently, it only functions as a comment.

Locked

If this option is checked, the machine/workplace has been (logically) deleted. In this case, the system does no longer permit the following changes:

- Order postings on the terminal
- Order postings on the MOC (e.g. using the "order overview" function)
- Changes when editing events

The graphic planning board of the Shop Floor Scheduling and the application *Workplace assignment* do no longer show the machine/workplace.

Blocked machines/workplaces are shown in evaluations and overviews. If blocked machines/workplaces are not shown, this is then described in the relevant documentation of the MOC application.

Tip: In applications where data is selected according to the responsibility area authorization, you can hide machines/workplaces if you remove the responsibility area.

Company

Use this field to differentiate the individual machines/ workplaces. The system can use this field for evaluation purposes.

Group

Use this field to assign the workplace/machine to a logical group. In planning, this is a capacity group. Capacity groups combine primary capacities.

If you create a new workplace, it is automatically assigned to a group of the same name (menu BDE: Master data > Workplaces/machines > Groups), which is defined as a capacity group. If the capacity group does not yet exist, the system automatically creates a capacity group and assigns the workplace.

Category

Enter the category of the machine. By means of this, you can enable a validation check according to the BDE configuration: *Master data > Order configuration > Order types*, tab *validation*, option *Check planned workplace/group/category on OP logon* (value *category*).

Year model

Enter a valid year model . The times to be posted are compared with this shift model when they are recorded. If you have not defined a planned year model in the HLS tab, the shift model entered here is also used in the Shop Floor Scheduling.

Standard rate, machine

Enter the arithmetical standard rate of machines for calculations. The Shop Floor Scheduling uses this value for some (evaluated) KPIs.

Standard labor rate

Enter the arithmetical standard labor rate for calculations. The Shop Floor Scheduling uses this value for the KPI "Evaluated labor utilization".

Performance level

You can enter the performance level of the workplace/machine in percent in this field. The Shop Floor Scheduling and the evaluation of material requirements integrate this value when calculating the remaining run time.

Incentive wage indicator

Defines the type of calculation used for incentive wages. This option is mostly used in combination with the incentive wages based on formulas for customer-specific configurations. In addition, use the "incentive wage indicator" as selection criterion for the *wage type determination* to calculate incentive wages.

Leave this field empty, if you do not use the incentive wage module.

The *incentive wages indicator G=group calculation* has a special meaning. If this option is set for a workplace/machine, you have to assign a premium group every time you log on an order. You can do this either via

- the "assignment of premium groups" option of the product group *Incentive wages* or, optionally, via
- an additional field in the terminal dialog for the logon of orders. If no assignment is available, the system rejects the logon of the order by issuing a validation error.

Therefore, you may only assign the incentive wage indicator *G = Group calculation*, if the group premium conditions are met in the incentive wages calculation, as otherwise orders can no longer be logged on!

You can specify the meaning of the other incentive wage indicators according to your requirements while customizing the system.

File

You can assign a graphic to each machine/workplace. The *workplace overview* or the AIP shows this graphic, for example. The following image formats are supported: jpg, gif, tif, bmp, ico, emf, wmf.



In the path configuration, you must have configured the following:

- the path ID "MOCWPIMG" for the MOC or SMA
- the path ID "HYDRA" (also see) for the AIP. The file name length of graphic files is restricted to 12 characters (8.3 notation). Note for Linux installations: only use lower case letters for file names.

Maximum capacity (KG)

If a machine is configured as melting aggregate, define the maximum capacity in KG here.

Accuracy class, unit, etc.

Information fields in order to describe the accuracy. These fields are only available if Test Equipment Management (PMV-PPK or PMV-SVP) is licensed and the right "mdresgenh" is assigned.

Data collection

Display 3rd list

Use the options described here to show/enable a third list in the main view of a Windows terminal (CTWIN / AIP). You can switch between the respective terminal lists depending on the options set. The following settings are possible. Please note that the contents displayed in the lists depend on the product group in use:

- Input material (MPL): shows logged on input materials/ batches.
- Resources (WRM): shows logged on resources and tools.
- Staff (BDE): shows logged on staff.
- Output material (MPL): Produced output batches are displayed.

Show material/PRT list when OP is logged on

This option is only relevant in connection with the WRM module and the resources logged on to the Windows terminals (CTWIN / AIP).

If this option is set and you log on an OP, a specific login dialog opens. This dialog includes a list of components/production resources and tools. This list shows resources that meet at least one of the following requirements:

- the option "posting to terminal" is set in the resource type;
- the option "log on with OP" is set to "explicit logon" for the resource.
- the resource is a so-called "required resource" (option is set for the resource).

Please note: If the workplace is relevant for MPL, the list also shows material components.

Sequencing list

This option defines which operations are displayed in the sequencing list of the terminal. The following settings are available:

- S Basic setting. The system takes the value from the option of the same name in the *HYDRA basic settings*.
- M Pool of workplaces. The terminal sequencing list only shows the operations planned for the workplace.
- G Pool of workplaces and groups. The terminal sequencing list shows operations that are:
 - planned for the current workplace or
 - for another workplace of the group or
 - that are still located in the pool of groups.
- K Pool of workplaces and categories. The terminal sequencing list only shows the operations that are planned for workplaces of the selected category.
- H Group control. The terminal sequencing list shows the operations that are
 - planned for the current workplace or
 - for another workplace of the group.

Number of OPs in sequencing list

Enter the maximum number of operations that are to be displayed in the terminal sequencing list. Enter 0 if you want to show all operations.

Compulsory sequence

Use this option to specify if it is mandatory to log on the OPs in the planned sequence. The following parameters are permitted:

- N Disabled
- J Enabled

If the parameter is "enabled" and you log on an OP, the system checks whether the order backlog for this machine/workplace includes an OP that is planned for the same time or previous to this OP, but has not yet been started (i.e. status = V/prepared). If yes, the system rejects the logon of this OP.

Note: If you plan orders in the system using the *Order sequencing* (menu Production control → Production support → Order sequencing) and you configure the sequencing list with any other option than "M" (pool of workplaces) and you enable the *compulsory sequence*, this might lead to a combination that does not make sense.

Please note for the sequencing list:

- If the sequencing list includes operations that are in the status "interrupted", you can log on these OPs at any time, irrespective of the specified compulsory sequence.

Dialog control

To meet this requirement, define a dialog control that deviates from the standard behavior for the workplace in the dynamic dialog configuration of the Windows terminal (CTWIN / AIP). Then refer to the dialog control in the dialog.

Use this configuration only as part of customizing the HYDRA system. Otherwise the configuration is not relevant.

Logon of several OPs

Select this option, if several different operations should be processed on the machine. Otherwise, the system only allows one operation to be logged on to the machine.

Possible values:

Y Log on as many OPs as required at the same time.

Please note: The system allows a maximum of 20 operations to be logged on simultaneously to a machine, if the machine is assigned to a terminal with operation mode *MDE*. If more than 20 operations must be logged on at the same time, MPDV must review the conditions in order to remove the limitation. If MPDV agrees to remove the limitation, you can do so, otherwise search for alternative solutions. MPDV analyzes the conditions as part of a service.

N You can log on one OP only.

1...9 You can log on a maximum of n OPs.

Posting

Quantity posting to staff

Use this function to post the quantity of order interruptions/logoffs to the person who is logged on for the longest period.

Detailed information about quantity posting to staff can be found .

Posting for OPs that are not logged on

Use this option if you want to

- interrupt
- finish
- report part quantities for operations that are not logged on to this workplace.



If you record quantities for an operation that is not logged on, the system posts these quantities onto the operation in the BDE module. The MDE module does not post the quantities.

If you want to use this function with the AIP terminal, the BDE posting dialogs that are installed by default require the following:

- use the simplified BDE posting dialogs (the so-called "") or
- customize the dialogs.

Then you will be able to enter an operation that is not logged on.

Posting of machine time with simultaneously logged on operations

If this option is set and OPs are logged on simultaneously, the system posts the machine time proportionately onto the operations.

- | | |
|---|---|
| Y | Proportionate posting on OP according to the number of OPs |
| N | No proportionate posting. If the option is not set, the complete machine time is posted for each operation. |
| V | According to the default quantity of the OPs. Make sure that the default quantity (target quantity in primary quantity unit) of the operation is > 0. |
| Z | According to the standard time of the OPs. Make sure that the standard time (processing time) of the operation is > 0. |

Please note:

This option is also evaluated for group workplaces and in general you should better not use this option for group workplaces.

Automatic logoff of staff when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

Use this option to configure the personnel-related data collection at MDE workplaces. If you use HYDRA MDE, the terminals can generate fully automatic shift ends. You can configure here if

- the staff logged on to the workplace should be logged off automatically at the end of the shift or
- if they should remain logged on.

- | | |
|---|--|
| Y | Always log off staff when the shift ends. |
| N | Always save staff when the shift ends except for manual logoff. |
| X | Evaluate the person's settings. The system searches for the corresponding settings of the person . |

Automatic OP posting when shift ends

This option is only relevant, if you set an "X" for (enable) the option of the same name in the order type.

- Y Interrupt and log on again at beginning of shift
- N Interrupt

Shop Floor Scheduling

Find further information about the HLS product group in the relevant HLS documentation.

Planning function

This option specifies whether a workplace or a machine will be displayed and if so, in which planning function.

- P Planning in the graphic planning board of the Shop Floor Scheduling or in the graphic order sequencing (GAV), i.e. you plan the workplace via the Shop Floor Scheduling or the graphic order sequencing; the workplace is then displayed in these applications, but not in the tabular order sequencing (AVG).

Note: There are also other settings that specify whether a workplace is displayed in the Shop Floor Scheduling or in the graphic order sequencing:

- the workplace must be assigned to a group identified as a "capacity group"
- you must be authorized for the responsibility area of this workplace
- planning profile

- H Only relevant, if you use the HYDRA Shop Floor Scheduling module (HLS).
Like P.

- T Reserved

- A Planning in the tabular order sequencing (AVG), i.e. you plan the workplace using the AVG product group.

- N No planning; the tabular order sequencing (AVG), the graphic order sequencing and the HLS module do not show the workplace.

Planned year model

Here, you can enter a special year model only used for planning in the Shop Floor Scheduling. This year model does not affect data collection and posting in the product groups BDE/MDE. If you do not define a planned year model, the system uses the year model (*Master data* tab) for the planning.

Availability

Define the available capacity of a workplace/machine. The default value for the available capacity is 1000 [per mill].

In the Shop Floor Scheduling, the capacity check and automatic assignment assume that each operation has a capacity requirement of 1000 [per mill], i.e. exactly one operation can run on the workplace/machine at a time. In case of a manual multiple assignment, a dialog informs you about the double assignment. If you use the automatic assignment, multiple assignments are generally not feasible.

Use this setting to extend the availability of the workplace such that a multiple assignment is permitted. If the workplace capacity allows, for example, processing of two operations at the same time, set the available capacity to 2000 [per mill] in this field.

If nothing is entered in this field or if you enter the value 0, the system interprets this as the default value of 1000 [per mill].

This functions requires a corresponding license.

Check personnel availability

Choose from the following options:

- Check if at least one person is planned
- Check personnel availability
- Check personnel availability and qualification

When you operations in the , the system checks whether persons are planned in the application for the time of the scheduling You will find further information on the display of personnel capacities in the Graphic Planning .



This option is only available if you enable the extension .

MPL

For further information on the MPL product group, refer to the relevant MPL documentation.

Batch management

Activates the entry of the batch number for this machine within the terminal posting dialogs.

Possible values are:

- N No batch processing
- L Batch tracing (input/ output batches) as part of HYDRA MPL/TRT
- D Throughput batch processing as part of HYDRA MPL/TRT
- J Individual batch tracing (CHV)

The following functions are only available in connection with the product group Material and production logistics and are supported only by Windows terminals (CTWIN / AIP).

Preceding material buffer

Irrelevant.

Subsequent material buffer

If you specify a material buffer in this field, the field *Target buffer* in each of the entry dialogs (e.g. output batch change, log off operation) is automatically populated with this value.

If you do not enter a material buffer in the input dialog (e.g. deleted from the input field), the system automatically posts the output batch to the material buffer specified in the "subsequent material buffer" field.

Automatic generation of batch number

If you set this option, the system automatically generates a batch number for the output batch to be produced. Otherwise, the system expects you to enter the batch number for the new output batch to be produced, when you log on an operation or change the output batch.

Please note: If, in the field *Batch management* you set the option D (= *Throughput batch recording*), the system automatically sets the value for the *Automatic generation of batch number* to "J". In this case, you cannot enter the batch number manually.

Consumption balance

When you log off an OP, the system opens an additional dialog (V_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. This option is only activated, once you have enabled the consumption balance for the material type of the output material.

Generate transport order for output batches

This option creates a transport order relating to batches for a generated output batch. The transport starts from the material buffer where the output batch is included. The configurations of the material type override the corresponding options of the resource configuration.

Generate transport order for input material

This option creates an article-related transport order relating to a material component, when you plan an operation for a machine via the Shop Floor Scheduling module. Transport starts from the output material buffer of the preceding operation. The configurations of the material type override the corresponding options of the resource configuration.

Quantities tab

This tab is only available if you select a resource of the type "MNR".

Conversion factors for base quantity

At the machine or workplace, you can collect the quantities in different quantity types and for different quantity accounts. In general, the system supports the following **quantity accounts**:

Yield

Scrap

Rework (Windows terminal CTWIN/AIP only)

Open quantity (problem quantity; Windows terminal CTWIN/AIP only)

The following **quantity types** are supported with each quantity account:

Primary quantity

Secondary quantity (Windows terminal CTWIN/AIP only)

Tertiary quantity (Windows terminal CTWIN/AIP only)

Basic quantity (Windows terminal CTWIN/AIP only)

The system design specifies the use of several quantity types or accounts. For example: If you want to enter the rework quantity manually, a corresponding input field must be configured in the input dialog (customization).

Use the quantity type "primary quantity" if you want to collect quantities automatically.

Quantity units and conversion factors for base quantity

Define a quantity unit for each quantity type. Use the alternative quantity accounts to enter data/quantities manually. In this case, the system does not convert quantities automatically.

If you do not enter data manually in the alternative quantity accounts, the server converts the quantities into the alternative accounts using:

- the conversion factors or
- the units that are configured in the MOC machine master data.



For further information on the conversion of quantities and examples, refer to the document

Basis for HYDRA-MDE quantity conversion

Define the basis for the quantity conversion.

- A Use the conversion factors of the OP that is logged on. If no operation is logged on, the system uses the quantity conversion stated in the machine/workplace configuration.
- M Use conversion factors from the workplace configuration for the quantity

conversion.

Units and conversion factors for base quantity (P)

Quantity unit (P)

Indicate the quantity unit you want to use for data collection at this machine/ workplace. If you collect quantities automatically, these quantities are generally primary quantities.

If you want to convert quantities automatically into another quantity type, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (S)

Quantity unit (S)

Indicate the secondary quantity unit you want to use for posting the quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity (T)

Quantity unit (T)

Indicate the tertiary quantity unit you want to use for posting quantities to the workplace/machine. If you want to convert quantities automatically, indicate the conversion factors for the base quantity here.

Units and conversion factors for base quantity

Quantity unit (B)

Indicate the base quantity unit you want to use for posting quantities to the workplace/machine.

Manual entry of quantities, yield

Manual entry of yield

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of yield

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting yield as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, scrap**Manual entry of scrap**

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of scrap

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting scrap as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, rework

Manual entry of rework quantity

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of rework

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting the rework quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

Manual entry of quantities, open quantity

Manual entry of open quantity

Set this option, if you want

- to collect quantities manually;
- to set off the quantities against another account;
- to post the manual quantities as cycles.

For Windows terminals this option does not affect the quantity fields displayed in the input dialogs. Change these quantity fields via dialog configurations (terminal configuration or customizing of the dynamic dialogs).

Allocation of open quantity

Requirement: Set the option "Manual entry".

Use this option to offset manually entered quantities against other quantity accounts. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.

Note

Do NOT set this option for DOS terminals, if in the counter configuration yield is offset against scrap or scrap is offset against yield.

Posting open quantity as cycles

Requirement: Set the option "Manual entry".

If this option is set, the system also posts manually entered quantities as cycles. Note here that the entered quantity is posted directly as cycles (partitioning is not integrated).

"MDE configuration" tab

This tab is only available if you select a resource of the type "MNR".

Monitoring

Monitoring type

Choose from the following monitoring types:

Monitoring via operating signal

No monitoring

Cyclic monitoring

If you select **cyclic** or **operating signal** monitoring, you can only enter a malfunction if the terminal prompts you to do so ("Assign malfunction"). If you do **not** use **automatic monitoring**, you can enter a new machine status at any time.

If you use the **cyclic** monitoring option, the machine automatically switches to the "production" status when counting pulses occur. If you select the "operating signal" option, the machine automatically switches to the status "production" as soon as the operating signal is set. If you do not use the "automatic monitoring" option, you must assign the "Production" status manually.

Entry of disturbance reason required with specified delay time in [s]



You can only use this function, if the following requirements are met:

- it is a Windows terminal (CTWIN, AIP)
- The Process Communication Controller (PCC) does not run in stand-alone mode.

If the system identifies a downtime without a reason, the terminal opens the input dialog "Change machine status" after the specified delay time. If the terminal goes back into production, the window still remains open.

If you now enter a machine status (during production), this data input activates a transfer posting event that changes the most recently recorded status from "General disturbance" to the newly entered status. If this change is ok, the window closes; otherwise, it remains open.

However, if the system identifies the next downtime (with or without a reason), you can no longer change to the previously noted status. The window closes automatically.

If the system identifies another downtime without a reason and the delay time has expired, then the input window opens as described above.

If the system identifies a downtime without a reason and the machine switches to production before the delay time expires, then the terminal does not automatically prompt you to enter a malfunction reason.

Important note:

This change only affects the HYDRA Machine Data Collection. The system does not correct the resource performance accounts of the currently running OP online!

Please note for data maintenance:

The tabular event maintenance of the MOC shows all changed machine statuses. However, you cannot edit the transfer posting event as it is locked. In order to perform recalculations correctly with respect to orders and machines, change the original event with the status "NOT ASSIGNED" to the correct status. The transfer posting event does not affect recalculation!

Minimum malfunction time

Specify a time in seconds for the minimum malfunction time. This value defines the time that a malfunction/disturbance must continue before the machine changes from the status "Production" to the status "Not assigned".

If operating signals are monitored, the status is directly changed. You can use the following explicit option in the MDEB2.ini to disable this behavior (deactivation of direct status change). Result: the status is only changed when the **minimum disturbance time** has expired:

MDEB2.INI
<pre>[INIT] ;Activating the direct status change (globally or for a specific machine) SetMStatusDirect=1 SetMStatusDirect@<machine number>=1 ;Deactivating the direct status change (globally or for a specific machine) SetMStatusDirect=0 SetMStatusDirect@<machine number>=0</pre>

Minimum cycle time

If you select the **cyclic** monitoring option, specify a minimum cycle time in seconds in this field.

The terminal uses this minimum cycle time and the target cycle that is stored with the (logged in) operation and that is set off against the cycle extension to calculate the maximum value. The terminal uses this maximum value as the default cycle time.

If both, the minimum cycle time and the target cycle stored for the operation, are 0, the default cycle time is set to 60000 seconds [per 1000 machine clocks].

Cycle extension

If you select the **cyclic** monitoring option, enter the percentage for extending the target cycle time in this field. Enter a value ranging between 0 and 5000.

The system offsets the target cycle stored with the (logged in) operation against this percentage. A value less than 100 is a shortened cycle; a value greater than 100 is an extended cycle.

Number of target cycles

If you select the "**cyclic** monitoring" option, enter the number of cycles (0 to a maximum of 9) after which the terminal automatically switches from a status unequal to "production" into the "production" status within the cycle time (requirement: the status that is unequal to production is not locked for the "production" status).

Some production processes provide machine cycles during the setup phase. Set a value greater than 0 in order to prevent the current machine status from changing immediately. Please note: The quantities you collect until the machine switches to the "production" status are neither posted as yield nor scrap.

Cycles to be evaluated

Reserved Enter 0 in this field.

Management

Posting during production lock

Use this setting to specify how to post the counting pulses that are collected while the status "production" is suspended. This configuration takes effect for all counters configured as "Yield".

Posting as scrap If this option is configured for the counter, the system offsets the counting pulses against the partitioning/ pulse factor and posts these pulses as scrap. Even if you defined another quantity account for offsetting, this one will not be used.

Posting as yield parts the system posts the counting pulses as yield

No posting the system does not post the quantities while the "production" status is suspended.

Pulse factor specific to machines

Use the pulse factor, for example, if you want to collect lengths (e.g. using a wheel).

Set the value to 0 for machines where a discrete or integral number of quantities (e.g. pieces) is collected per pulse. In this case, the pulse factor is not evaluated. That means, the number of cycles posted corresponds to the actual pulses transferred via the MSS (machine interface).

The MSS (machine interface) records the signals transferred from the machine (counting pulses). According to the configured number of pulses, the system calculates and posts the quantities as follows:

Quantity for the machine = pulse * partitioning for the machine/ pulse factor for the machine

Quantity for the operation = pulse * partitioning for the operation/ pulse factor for the operation

Please note: The pulse factor will be calculated as a fraction. When the quantity is calculated, the pulse is used as denominator and the partitioning is the numerator.

The system interprets pulses that occur during a malfunction or a production lock (configuration of *Posting during prod. lock > scrap*) as scrap. Also use the above-mentioned formula to calculate the scrap quantities.

Partitioning specific to machines

Enter the partitioning specific to the machine in this field. Multiply the machine-specific partitioning by the partitioning stored with the operation in order to integrate the machine-specific partitioning into quantity calculation. Enter the value 1 in this field, if you do not want this to happen.

Extended weekend automatic

If you select this option and the system is configured accordingly, the system assigns at the beginning of the shift the status that was available before status 999 was activated.

Note:

To use this option, the workplace must already be assigned to a terminal.

Find detailed information about the automatic activation of status 999 in the document .

Waiting period, short-term disturbance

Configure a *short-term disturbance* status for each machine/ workplace to improve the overview, e.g. in the machine history. Use this status as a "repository" for unconfirmed statuses, which only existed for a specific (short) period.

If the terminal automatically identifies a downtime and the machine automatically goes back into production, the system checks if this disturbance is shorter than the time period configured for short-term disturbances.

If this is the case, the still unfounded malfunction is justified with the status that is configured as the "short-term disturbance" status for the machine.

Inputs/ outputs

Machine lock/ Target quantity reached/ Machine downtime/ Free I/O

Enter the logical output where a digital signal should occur when the corresponding status is available.

Machine lock output	The system sets this output, if you enabled the option "set machine lock output" in the current machine status.
Target quantity reached output	The system sets this output, if the collected yield reaches the target quantity of the OP.
Machine downtime output	The system sets this output, if the machine is in a status unequal to <i>Production</i> . When changing to the <i>production</i> status, the system sets the output back to 0.
Free I/O	Free input/ output for customizations.

Use these statuses for connecting a monitoring light or a horn, for example.

Enter the corresponding number in one of the fields in order to assign an output and to specify which relay is interconnected by the terminal when the predefined status occurs. Enter "0" to prevent any action. Note that you cannot assign a terminal output more than once.

Please note

Specify the statuses that trigger the activation of the machine lock in the *Status assignment*.

Generally, enter the value "1" in the input field, when the machine lock is activated via the available relay output of a DS 100. In this case, the system sets the machine lock if

- a correspondingly configured status occurs and
- the status is not assigned.

Output batch change**

Customer-specific assignment of an input with an automatic output batch change (MPL). By default, enter 0 in this field.

PDE (Process Data Collection)

Collect process data

This parameter specifies if the system collects process data for this machine. If this parameter is not set for a machine, you cannot collect process data for this machine.

External connection



The AIP 8.2 and/or the PCC in stand-alone mode (MDE-Blade 2 Version 8.1.0.1) do no longer support the options marked with **. As they use other configurations for the connection.

External connection

If this machine is assigned to a master terminal the following connection options are available:

No external device	External devices are not connected
DS100	DS100 connection
Arburg control system**	Arburg connection
Engel interfacing**	Connection of Engel machines
MT3**	MT3 connection
PDE**	Process data collection

If you activate a DS100 or MT3** connection, you can select the field "device address". If you activate the option "Engel interfacing", you can select the field "serial number". If you activate the option "Arburg server system", you can select the field "class".

Note regarding the combination of connections on a master terminal:

"DS 100" and "No external device": allowed

"MT 3" and "No external device": allowed

"MT3" and "DS 100" not allowed!

Serial number (Engel interfacing)**

Enter the serial number of the connected Engel machine. Set the option "EMS machine interface" in the HYDRA basic parameter settings if you want to use Engel machines.

Device address

You can select this field, if you activate a DS100 or MT3** connection. Enter the device address of the sub-bus participant.

"Resource configuration" tab



For resources of type "MNR", only the fields marked with "*" are available:

- Family (section *resource master data*)
- Cycles (section *target utilization*)
- Runtime (section *target utilization*)

Resource master data

Type

Identifies the type of resource:

Resource: A resource can be uniquely identified, i.e. the resource is actually present. Its quantity is always 1.

Anonymous resource: An anonymous resource cannot be uniquely identified. If the identifier is set, then you can change the value in the field *Number* from 1 to another positive integer value. You cannot post data onto anonymous resources because anonymous resources do not relate to one specific resource.

Required resource: A required resource stands for one or more actual resources that can be identified. Specify in the configuration *WRM: Master data > Required resources* which resources are represented by a required resource. The number results from the number of actual resources assigned to the required resource.

Please note: If this field is empty, the resource is implicitly an ("actual") resource.

Equal type

Reserved for future modifications.

Version

Revision number; store here the program version for resources of the type DNC.

Quantity

You can only edit this field, if it contains an anonymous resource and the option *Anonymous resource* is set (see above). A value > 1 indicates how many of these resources are available.

This field is calculated automatically for required resources.

Family*

Assign a resource family. If you change the resource family subsequently, an information dialog appears as a warning because user fields might possibly be assigned via the resource family.

Target utilization

Cycles*

The field *Cycles* provides additional information. The cycles value defines how long the resource is to be used.

Runtime*

The field *Runtime* provides additional information. It defines how long the resource is to be used.

Input unit

Input unit

Absolute value limit (EMG 8.1, function authorization: resablim)

Enter the absolute value limit of the (meter) resource. The energy monitor shows this limit value in addition to the current meter reading. Use the Escalation Management to generate an escalation message, if the counter value of the resource exceeds the specified absolute value limit. You need the function authorization "resablim" to view this field.

Actual utilization

The periods when a resource was logged on to a workplace are the basis for posting the *cycles (clocks)*, *runtime*, *yield*, and *scrap* as *actual utilization*.

Clocks

The cycles (clocks) posted for the resource up to now.

Runtime

The total time in hours posted for the resource up to now. The total time is the sum total of all times posted onto RPA 1 to 11.

Yield (B)

The yield posted for the resource up to now (base quantity unit).

Yield (P)

The yield posted for the resource up to now (primary quantity unit).

Scrap (B)

The scrap posted for the resource up to now (base quantity unit).

Scrap (P)

The scrap posted for the resource up to now (primary quantity unit).

Configuration

Target cycle

Target duration in seconds for 1000 machine cycles if this tool is used.

Please note: The target cycle stored in the OP is relevant for the planning in the HLS module and for the machine data collection at the terminal.

Original partitioning

Partitioning of the tool (= number of cavities) when using this tool.

Current partitioning

Current partitioning of the tool. This value can deviate from the original partitioning, e.g. if the original quantity can no longer be produced with one cycle/clock due to a tool defect.

Always use the current partitioning to post cycles to the tool.

Please note: The partitioning stored in the OP is relevant for the planning in the HLS module and for the machine data collection via the terminal.

Partitioning due to cavities

If you set the option "partitioning due to cavities", the system (re-)calculates the fields "current partitioning" and "original partitioning" using the values defined in the cavity management. Then, you can no longer change the fields manually.

Log on with OP

Use this option to specify whether or not you want to log on the resource with the OP. To do so, the resource must be included as a component in the operation's list of production resources and tools.

Possible values:

None: The resource is not logged on.

Implicit: The system automatically (implicitly) logs on the resource that is assigned to the operation as a production resource and tool; you can neither log on the resource manually (explicitly) nor change the logon.

Explicit: You can manually (explicitly) log on the resource that is assigned to the operation as a production resource and tool or you can log on another resource instead. If you do not log on the resource or another resource explicitly, the system implicitly (automatically) logs on the current resource; in this way, the current resource serves as a "default".

Please note:

If you log on another resource explicitly (manually), this resource will be logged on for the resource that has the same *resource type* in the operation's list of production resources and tools. For this reason, you can only log on those resources explicitly (manually) that are included as a requirement in the operation's list of production resources and tools. In this way, you cannot log on a resource that is not included as a requirement in the list of production resources and tools (the resource must be entered in the list).

In general, you should not enable this option for the resource type DNC. The DNC product group handles this differently (NC programs are logged on separately).

The system also logs on resources that are defined in the BOM of the machine.

Parallel logon/ planning possible

You can log on/plan the tool simultaneously.

Please note: You can only log on a resource to one machine more than once. Consequently, the option "Parallel logon possible" refers to several different OPs logged on to one machine.

In this case, the system posts data proportionately as follows:

- Post quantities proportionally.
- Post times 100% for each resource. This means that the system posts double the time to the resource, if the resource is logged on twice.

Post to resource

Specifies whether or not the quantities and times are posted to the resource. Due to a high degree of complexity, you should only assign this option to those resources that you actually want to evaluate.

Collective lock

If you lock a lower-level (assigned) resource using the BOM function, the system sets a collective status for the higher-level resource. If this collective status is set, the system treats the higher-level resource as locked when a download request is made.

If you enable this function, the system passes the collective lock to the higher-level resource.

Planning**Setup time**

Duration in hours for setting up the tool.

Please note: The setup time stored in the OP is relevant for the planning in the HLS module.

Teardown/retooling time

Duration in hours for removing the tool.

Please note: The retooling time stored in the OP is relevant for the planning in the HLS module.

Assignment

Not used. The system uses the configuration option of the same name stored in the resource type to integrate the resource allocation in the HYDRA Shop Floor Scheduling.

Evaluation**Integrate in evaluations**

Reserved for future modifications.

File**File exists**

Shows whether or not the file is stored in the specified path. A cyclic process checks the files and sets the options subject to whether or not the file is available.

File name

File name; without file extension for DNC. The system adds the file extension according to the configuration in the resource type. The defined paths specify the storage location.

Comparison resources

Enter two comparison resources for energy consumption resources. They will then be shown in comparative evaluations/reports, e.g. the energy monitor.

Resource 1

Resource number of the resource to be compared.

Resource type 1

Resource type of the resource to be compared.

Resource 2

Resource number of the resource to be compared.

Resource type 2

Resource type of the resource to be compared.

Accuracy

Enter more detailed information on measuring accuracy and measuring range for test equipment resources.

Tab *User fields*

You can use user fields to store additional customer-specific information in the MES. The *user fields* tab includes eight sub-index tabs, which each has eight additional user fields. The so-called user field key specifies the available user fields and their meaning.

The workplace and resource configuration provides data of two basic object types. You can also edit this data in the workplace and resource configuration: on the one hand these are machines and workplaces and on the other these are the resources. Machines and workplaces are also "resources". But resources are not automatically machines and workplaces.

Object type

The system configures the user fields of machines/workplaces in relation to the object type "MNR". The system stores data contents to the machines/workplaces table and the resources table of the database to ensure data consistency.

The system configures user fields for resources in relation to the object type matching the resource type of the resource (example: create resources of the type "PAC" in relation to the object type "PAC"). The system stores data contents to the the resources table of the database.

User field key

Each user field key describes a combination of user fields. The management of the user field key (and therefore the meaning of the fields) is different for each object.

User fields

The following user fields are available after configuration:

Field data type	Number of fields
Date	6
Numeric, time, duration	16
Decimal value	6
Text field, length 1	16

Field data type	Number of fields
Text field, length 10	6
Text field, length 20	14
Text field, length 40	2

Each page shows a maximum of 8 fields.

By default, no user field keys are defined. Configure the system accordingly to support this kind of user fields.



As the table shows resources of different types, use the user field key "SYSTEM" of the object "RES" to identify the column headings for the user fields.

Comment tab

Store additional resource comments in the "comment" tab.

Main tab Resource attributes

Shows additional resource attributes via the user field definitions of the resource family. Use the "resource attributes" button for editing.

Main tab Resource list

Shows the resource list for the selected resource. Click the "resource list" button to go directly to the BOM application for editing purposes.

Main tab DNC versions (available as of DNC 8.2)

Shows the available versions of a DNC resource including a flag indicating the currently applicable version. HYDRA provides this valid version for machine downloads.

Toolbar

General tab



Insert

Function authorization: mdres.create

Opens the dialog for adding a resource. This dialog provides the fields that match the selected resource type.



Copy

Function authorization: mdres.copy

Opens the dialog for copying an existing resource. Subject to the selected resource and its resource type, the copy function differentiates the following:

- Copy function for resources of **resource type = MNR** (workplaces, machines)
- Copy function for resources that do **not** have the **type MNR**

Copy function for resources of resource type = MNR (workplaces, machines)

From: resource type, resource, short name, name

- Resource type (fixed "MNR")
- Workplace/machine number
- Short name
- Name

of the workplace you want to copy. You cannot change these values. They derive from the selected data record.

To: resource type, resource, short name, name

- Resource type (corresponds to the resource type of the workplace you want to copy; cannot be changed).
- Workplace/machine number
- Short name
- Name

of the target workplace.

Copy machine status

Function authorization: mdmst.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy counter configuration

Function authorization: mdctr.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Note that the counter numbers of the new workplace are identical with the counter numbers of the workplace you copied. If necessary, you have to adjust the counter numbers.

Copy reasons

Function authorization: mdreas.copy

If you set this option, the system automatically creates and transfers all of the workplace you want to copy to the new workplace.

Copy function for resources that do not have the resource type MNR

The copy function for all resources that do not have the type MNR opens the "insert" dialog and takes over the details from the previously selected resource. But you can edit and change all fields.



Edit

Function authorization: mdres.edit

Opens the dialog to edit a resource and provides the tabs and fields of the relevant resource type.

As of MES Weaver 4.0_{pe}, you can change master data of several selected resources of the same resource type at the same time. You can select up to 10 fields and assign a value. You require the function authorization **mdresmm** to edit several resources at once.



Delete

Function authorization: mdres.delete

Deletes one or several selected resources.

Resource tab



Configuration – resource status

Opens the application "resource status" to define statuses for all resources that do not have the type MNR.



File - show file

Opens the file view – only available for document resources, which are configured as file-based resources without DNC processing in the *Resource type*. And only available if the relevant license and function authorization are available.



Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Go to – required resources

Opens the "required resources" application. The selected resource is entered as default value for the required resource.



Go to – cavity assignment

Opens the "cavity assignment" application. The selected resource is entered as default value.



Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.



Functions – Measures

Opens the *Measures* application.



Functions – Status change

Opens the dialog to change a resource status. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.



Functions – Release of resource

Opens the dialog to release a resource. The checkbox *Including subordinate resources* is not relevant and reserved for future extensions.



Functions – Stock transfer

Opens the dialog to transfer/relocate a resource.

Workplace tab



Configuration – status assignment

Opens the application "status assignment". The system enters the selected resource in the corresponding field.



Configuration – counter configuration

Opens the application "counter configuration". The system enters the selected resource in the corresponding field.



Configuration – terminal assignment

Opens the application "terminal assignment". The system enters the selected resource in the corresponding field.



Entry – reasons

Opens the application "reasons". The system enters the selected resource in the corresponding field.



Entry – Operator positions

Opens the application "operator positions". The system enters the selected resource in the corresponding field.



Entry – premium indicator

Opens the application "premium indicator". The system enters the selected resource in the corresponding field.



Groups - groups

Opens the application "groups". The system enters the group of the selected resource.



Groups – group assignment

Opens the application "group assignment". The system enters the selected resource in the corresponding field.



Miscellaneous – cycle parameter

Opens the application "cycle parameter". The system enters the selected resource in the corresponding field.



Miscellaneous - workforce requirements of workplaces

Opens the application "workforce requirements of workplaces". The system enters the selected resource in the corresponding field.

DNC tab

The tab is only available, if you select a DNC resource. These are resources configured as resources with DNC processing in the resource type.



Configuration – resource status

Opens the "resource status" application.



Configuration - assignment of DNC family to machine

Opens the application "assignment of DNC family to machine".



Copy resource attributes (as of DNC 8.2)

Copies values of resources attributes from one resource to another. Both resources must use the same user field key.



File - comparison editor

Opens the comparison editor for the selected resource or resources. See [below](#) for further information.



File - export

Exports the file specified for the resource. You use the file explorer to specify the target file.



File - import

Imports the file specified for the resource. You use the file explorer to specify the source file.



File - viewer

Opens the file specified for the resource using the defined viewer program.



File - editor

Opens the file specified for the resource for editing using the defined editing program.



Set valid version (as of DNC 8.2)

Only active, if you select a version in the *DNC versions* tab. The selected version is set as the new and valid version.



Go to - resource attributes

Opens the application "resource attributes". The selected resource is entered as default value.



Go to - resource list

Opens the *Resource list* application. The selected resource is entered as default value for the higher-level resource.



Functions – Status change

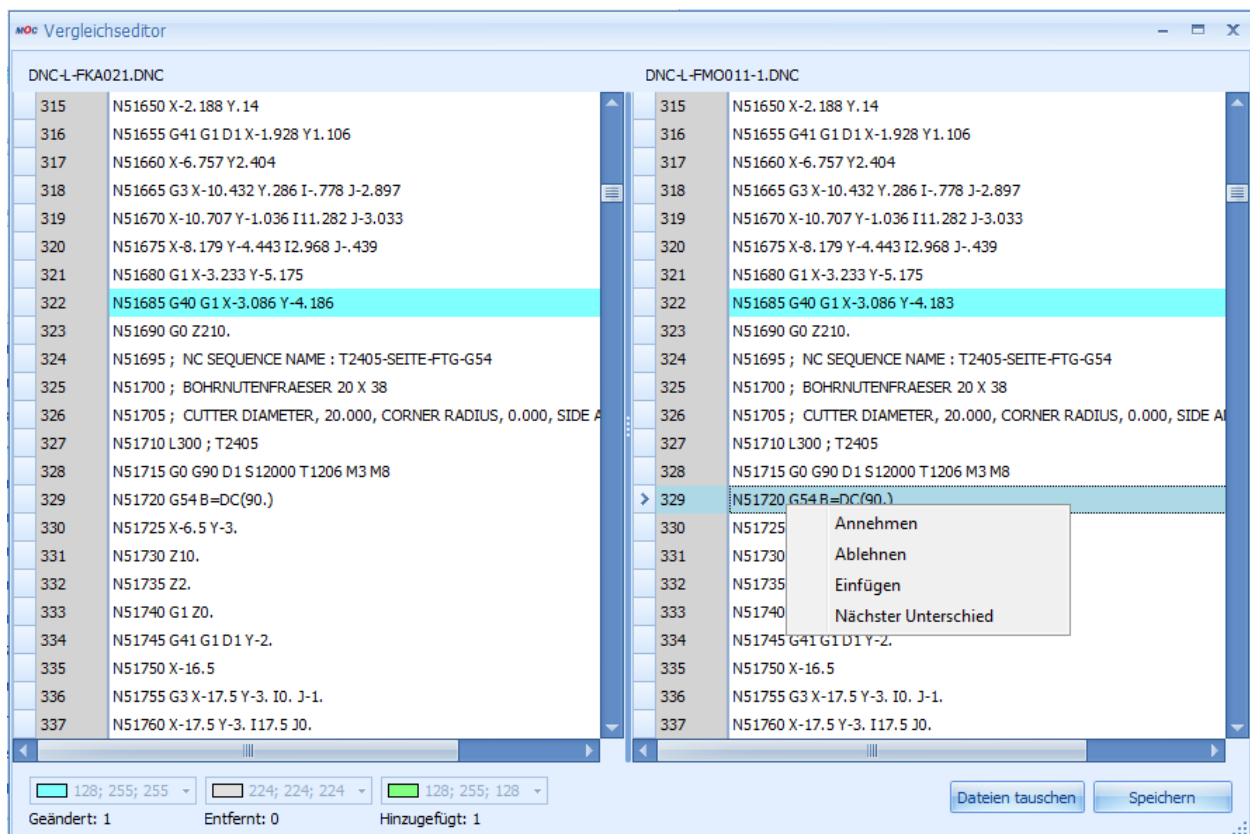
Opens the dialog to change a resource status.



Functions – Release of resource

Opens the dialog to release a resource.

How to use the comparison editor



The comparison editor compares the files attached to the DNC resources. Two operation modes are available:

Selection of one resource:

The editor shows the released resource and the optimized version of the resource for comparison. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor. You can only use this mode for DNC types with the file processing type "optimized".

Selection of two resources:

If you select two resources before you open the comparison editor, the editor compares the two selected resources. You can select the file type. You can change the file displayed on the right-hand side of the editor. Once you have made the changes, the comparison editor transfers these changes to the system, like the simple editor.

Click the relevant buttons or use the context menu (right clicking) to start the functions of the comparison editor:

- Reject: Rejects the difference identified (on the right). Accepts the value from the left file. The editor does no longer highlight the difference.
- Keep: Accepts the difference identified (on the right). The editor does no longer highlight the difference.
- Next difference: Goes to the next difference.
- Insert: Inserts a row at the current position.
- You can always change the contents of a row. Click the row and enter a value. Press ESC to quit the row without changes. The editor then highlights the row as "changed".
- Swap windows: Click this button to swap the windows. This function is necessary if you compare two resources. The place where a resource is displayed results from the display order in the table; the system does not know, which resource must be changed. If you only select one resource, this button is not available as in this case you can only change the optimized program version.
- Save: Saves the changes made to the file on the left-hand side.

Processing notes for workplaces and machines

Configuration changes

Restart the terminal which the workplace/machine is assigned to in order for the terminal program to interpret the configurations or modifications made to this workplace/machine.

Deleting a machine/ workplace

In a first step, the system shows a confirmation prompt asking if you really want to delete the machine. If you confirm this prompt, the system makes an attempt to delete the workplace. You can only delete a workplace successfully, if:

- you have not yet collected data for the workplace;
- you have currently not assigned the workplace to a terminal or a line;
- you have currently not logged on operations to the workplace;
- you have not planned operations for the workplace.

If you delete the workplace successfully, the system also deletes all configuration data, e.g. status assignments, for this workplace.

Checking Business Parameter Containers (BSCs)

See [for further details on how to check the system against business parameters.](#)

3 Resource List

Summary

Menu	Production Facility Management → Production Facility Administration → Resource List
Transaction code	resbom
Function authorization	mdrbom

HYDRA offers the option to map so-called resource lists. Further resources that can be used as subresources can be added to every resource. The BOMs can be used in many applications with regard to resource management.

Usage

In general, only one resource per resource type is stored in the production plan in the PPS system; it is transferred via the interface. However, because resources and, particularly in this case, *Tool* and *DNC* type resources, often have a more complex composition, the relationships between resources must be able to be mapped.

For this reason, HYDRA offers the option to map so-called resource lists. Further resources that can be used as subresources can be added to every resource. The superordinate resource, also called a "handle" or a "packet", is the overall term for other required resources that are needed in an operation. The number of the "handle" provides a unique specification for the entire resource (there is precisely one existing instance). It does not make any difference if a "handle" or a "packet" only has resources of the same resource type or the same resource family in the resource list.

Examples for resources with BOMs could include:

Base framework with exchangeable parts

Combination tool consisting of several components

NC packets with single programs and/ or with NC packets

Please note

If the superordinate resource is not a real, existing resource, then it is called a virtual resource. Both the virtual resource and its components can be planned and/ or posted if they can be uniquely identified.

Integration

Any resources can be connected with each other in hierarchical structure using the BOMs. In connection with the ERP interface and the production resources and tools stored for the operation, the resources can be managed automatically with the collection of the order data.

Requirement

As a basis for the BOM function, the WRM-MGM basic packet must be present along with the license for the WRM-WRP resource packets

Selection criteria

The following selection criteria are available in the application:

Resource type

Type of resource.

Resource family

Family to which the resource is assigned.

Resource

Unique resource identification.

Designation

Designation of the resource.

Field descriptions

Resource type

Resource type to which the resource is assigned.

Resource

Superordinate or assigned resource.

Number

This value describes the number of assigned resources. It is considered in the assignment of the resource and the assignment check in the HYDRA shop floor scheduling (only resources with resource types configured with *Assignment* = "G"; not with file-based resources).

Position

In the assignment it is required that a BOM item is input. The position is used for the defined display sequence in lists and in the hierarchical display of the BOM. The position must be unique within a superordinate resource.



A maximum of 5 levels are supported in BOMs .

When copying is performed, the maintenance dialog is called in which the values of the currently selected entries are the default values. This makes it easier to create further assignments.

Toolbar



Superordinate resource information

Calls the superordinate resource information



Subordinate resource information

Calls the subordinate resource information

4 Resource types

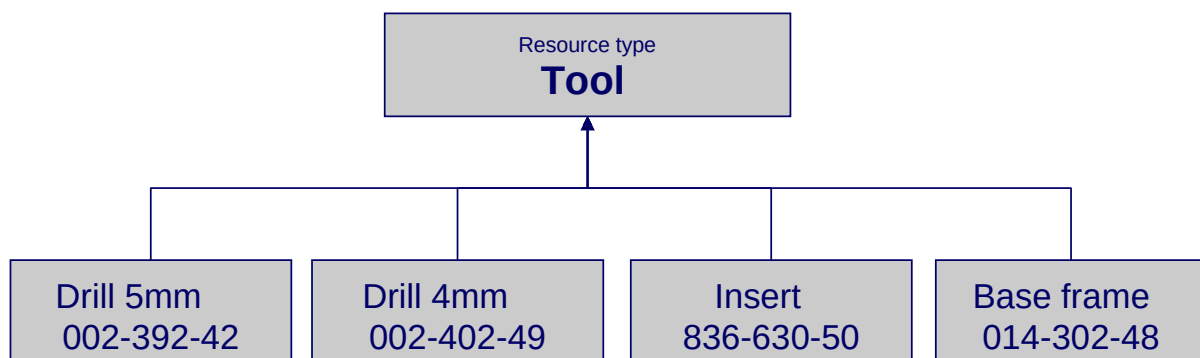
Overview

HYDRA menu	Master Data → Resources → Resource types
FEDRA menu	Detailed scheduling → Master data → Resource types
Transaction code	restyp
Function authorization	mdrtyp.*

This document describes the application "Resource types" on the client.

Purpose

Resources are classified in resource types with respect to their function and use. For example, you can group tools by assigning them to the resource type "Tool".



You use resource types not only to classify resources, but also to control specific functionalities. For example, the resource type is used to control whether or not an assignment check for resources is made in the shop floor scheduling (only relevant if the additional function is used).

The resource types listed in the following table have been predefined by MPDV. They are created as part of the implementation process.

Resource type	Abbreviation/ Ident
Machine	MNR
Tool	WNR
Staff	PER
Gage	PRM
Device	VOR
DNC-Programm	DNC
Document	DOC
Energy counter	ENE

Note

Various resource types are subject to certain technical restrictions. For example, users cannot delete the resource types "Machine", "Tool", and "Staff". Further information on this subject can be found in the chapter about configuring resources types.

Integration

You use resource types as a characteristic to specify differences between resource objects. The resource type therefore is the top classification criterion.

Selection parameters

In the selection panel, you can filter by higher-level or assigned resources. The application provides the following selection criteria:

Resource type

Type of resource

Field descriptions**ID**

Unique internal key.

This value may not be modified for the resource types delivered by MPDV because a range of processing depends on it.

Resource type

Unique "self-explanatory" designation of the resource type, e.g. "Machine" or "Tool".

You can select this value in the various functions. Only the resource type allows you to identify a resource or its resource ID uniquely. That is why, evaluations also show the resource type of the resource.

Description

Explanation of the resource type; in form of a comment.

User field key

Refers to a valid user field key

Field description for tab **General**

Assignment

This option specifies whether or not a resource of this resource type should be assigned. An assignment is a prerequisite for performing an availability check for the resource when planning an OP on a machine in the detailed scheduling of the HYDRA shop floor scheduling (HLS).

Possible values:

N = No, no assignment

G = Assignment of the total duration of an operation

Please note: For the resource type DNC, the setting should be set to *None* because there is an "endless" capacity for resources of this type.

Please note: For resources of type MNR (machines) the setting has no significance because machines are always assigned as primary capacities.

Automatic creation

Identifier that indicates whether or not a stock is to be created automatically for a resource of this type if this resource is transferred using the component list from the PPS and if this resource does not yet exist in the WRM product group.

Please note: This identifier is inactive and cannot be changed for the resource type "Machine".

→ Status assignment

This identifier specifies whether or not a status configuration (menu *WRM: Master data* → *Status assignment*) is allowed for this resource type.

Note:

This identifier is inactive and cannot be changed for the resource types "Machine" and "Staff".

Log on with OP

This identifier is used to control whether or not a resource of this type, which is assigned to the operation as a component, is logged on. Possible values:

None: The resource is not logged on.

Implicit: The system automatically (implicitly) logs on the resource that is assigned to the operation as a production resource and tool; you can neither log on the resource manually (explicitly) nor change the logon.

Explicit: You can manually (explicitly) log on the resource that is assigned to the operation as a production resource and tool or you can log on another resource instead. If you do not log on the resource or another resource explicitly, the system implicitly (automatically) logs on the current resource; in this way, the current resource serves as a "default".

Note:

This value is used as a "Copy template", if you manually create a resource for the first time in the MOC. Er wird direkt in das entsprechende Konfigurationsfeld der (neu angelegten) Ressource übernommen. For the rest of the process, only the value specified for the resource is used.

In general, this identifier should be inactive for resource type DNC because a specific processing exists for it in the HYDRA product group DNC (NC programs are logged on separately) (only applies when using HYDRA).

Post to resource

This identifier is used to specify if a resource can be posted to or not. If the identifier is set, the resource is logged on automatically with an operation logon.

The identifier must be set if cycles and times are to be posted for resources of this type, e.g. for evaluating the use of resources (in the evaluation function of the same name) or for consideration in the maintenance calendar (WRM-WWR).

This identifier should not be set for resources of type "Document", "Staff" and "DNC".

Note:

This identifier is only considered for resources that contain the number value 1 in the resource stock.

This value is used as copy template if you manually create a new resource on the client. It is directly transferred to the relevant configuration field of the (newly created) resource. For the rest of the process, only the value specified for the resource is used.

Consider in evaluations

Reserved

Posting on the terminal

If this option is set, resources of this resource type are displayed in the 3rd list.

Counter/energy resource (EMG 8.1)

Specifies if it is a counter or energy resource (only applies when using HYDRA).

Field description for the tab *Maintenance*

Maintenance monitoring based on the following RPAs

This field includes the information which operation hours of which RPAs are used as reference for the maintenance monitoring according to hours of operation.

Note:

This identifier is only relevant in connection with the additional function WMR-WTK (maintenance calendar).

Field description of the tab *DNC/Documents*

DNC processing (only applies when using HYDRA)

Specifies the behavior of HYDRA for DNC resources of this type:

K: No DNC processing (for resources of this type)

All resources except for DNC resources are configured with this processing option.

L: Local program

Generated by upload to the machine and saved in HYDRA.

E: External programming system

The file is located on an external system and from there it is processed through HYDRA.

O: Optimized program

Generated by upload on the terminal and then transferred to the external programming system through HYDRA.

R: Replacement procedure (**DNC 7.2**)

The upload overwrites the version that is applicable at that time.

V: Version based (**DNC 8.2**)

An upload of an existing resource generates a new version.

File-based

The programs are file-based.

Uploaded version set by default (**DNC 8.2**)

If this option is checked, the uploaded version is automatically set as valid version in version-based DNC processing.

File extension for valid programs:

The files for upload-download can be distinguished by their file extension. Only valid programs are used for downloads to the terminal.

File extension for optimized programs:

The files for upload-download can be distinguished by their file extension. New programs optimized by upload are provided with this extension. These programs must be "released" before being downloaded again by the programming system by changing the file extension.

Path

A path configured in HYDRA for saving the files in the server or programming system.

File extension for program description 1...3 (**DNC 7.2**):

A total of 3 other file extensions are available for saving descriptions, etc.

Field description for the tab "compensation"

Posting record after [hh:mm:ss] hours

Specifies the time when a compensation record is to be written at the latest.

Use cancellation documents for editing

If this option is enabled, cancellation records are created as part of the editing process.

5 Resource Families

Overview

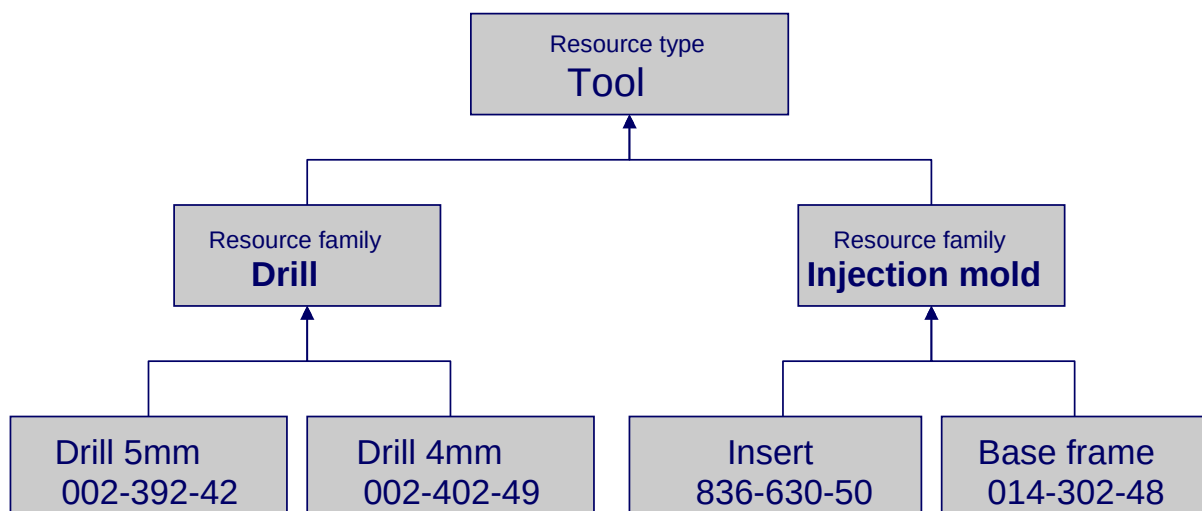
HYDRA menu	Master data → Resources → Resource families
FEDRA menu	Detailed Scheduling → Master data → Resource families
Transaction code	resfam.*
Function authorization	mdrfam

This document describes the application "Resource Families" on the client.

Purpose

If you look at the assignment of resources to resource types, you soon recognize that in a manufacturing company various resources of the same type exist that are possibly handled quite differently. This means that in general the classification by resource types is not sufficient to organize resources in a useful way.

If you define "resource families" (groups), you can introduce sub-classes of resource types. The diagram below illustrates how the resource type "Tool" is sub-divided into the two resource families "Drill" and "Injection mold". Each of the individual resources is assigned to one of the two resource families.



Integration

The resource families offer another structural level subordinate to the resource types. You can use resource types to define the master/detail user fields of resources. You can improve these master/detail user fields through definition in the resource families. In particular for DNC, you can use resource families as the main search criterion and assignment criterion for machines.

Selection parameters

In the selection panel, you can filter by superordinate or assigned resources. The application provides the following selection criteria:

Resource type

Type of resource.

Resource family

The resource family to which the resource is assigned.

Field descriptions

Resource type

Resource type to which the resource families refers.

Resource family

Unique, descriptive name of the resource family.

You can select this value in the various functions. Only the resource type allows you to identify a resource or its resource ID uniquely. That is why, evaluations also show the resource type of the resource.

Description

This field includes the description of the resource family; serves as a comment.

Responsibility area

Definition of the responsibility area. If you specify the responsibility area for a resource family, you also specify the responsibility area for the assigned resources. The responsibility area controls the visibility and editing options for these resources.

Field description for tab *General*

User field key

Reference to a valid user field key. The user field key entered here overwrites the entries in the resource type.

Note regarding DNC filtering using a DNC family and its search fields (when using HYDRA only):

The definition of suitable user field combinations is important if you want to use the flexible filter and search functions in the DNC module. You can define such user field combinations as part of the configuration. The user is responsible for the assignment and utilization of these user field keys. Use the defined search fields in the terminal to filter the DNC records in addition to the DNC family of the machine. You can also use these fields as search criteria in the MOC.

Starting with release DNC 7.2, the following preconfigured user field keys will be delivered:

User field key	Description of the search fields
DNC_K	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited
DNC_K_V	Plastic injection molding: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Tool, mandatory field, cannot be edited 4. Version, mandatory field, cannot be edited
DNC_K_W	Plastic (tool reference only): 1. Tool, mandatory field, cannot be edited
DNC_K_WV	Plastic (tool reference and version): 1. Tool, mandatory field, cannot be edited 2. Version, mandatory field, cannot be edited
DNC_NC	NC programs: 1. Article, mandatory field, cannot be edited
DNC_NC_V	NC programs: 1. Article, mandatory field, cannot be edited 2. Version, mandatory field, cannot be edited
DNC_NC_M	NC programs: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited
DNC_NCMV	NC programs: 1. Machine, mandatory field, cannot be edited 2. Article, mandatory field, cannot be edited 3. Version, mandatory field, cannot be edited
DNC_FREI	1. Search field 1, Text20, mandatory field, can be edited 2. Search field 2, Text20, optional field, can be edited 3. Search field 3, Text20, optional field, can be edited 4. Search field 4, Text20, optional field, can be edited

Notes on the DNC administration

DNC records are used exclusively with machines. In order to avoid false entries or false allocations, every machine is assigned to a definite DNC resource family. This is stored in the machine resource data (the *Resource family DNC* field). In this way, you can make sure that only programs belonging to a particular resource family and, indirectly, to a particular resource type can be loaded to a machine.

Furthermore, for the management of DNC records certain criteria are necessary, which, among other things, simplify selection and evaluation, thereby simplifying location and editing and enabling inspections. As widely different machine types can be dealt with by DNC administration (including, for example, injection mold machines, printers and NC machines), a rigid determination of these criteria is not advisable. For this reason, the resource family exists. You can use the user fields to assign attributes to the resource families. These attributes describe and specify the variable parameters.

Therefore, you can use the attributes for identification purposes and you can assign validation functions and allocations. In doing so, you establish a connection between the DNC programs on the one hand and the machines and operations on the other (see section entitled "User fields").

There are variables, such as the temperature and humidity, which influence the behavior of the machines and can therefore have an influence on production. You can also record these "environmental factors". For this purpose, you just have to define further attributes in the user fields.

6 Resource status

Overview

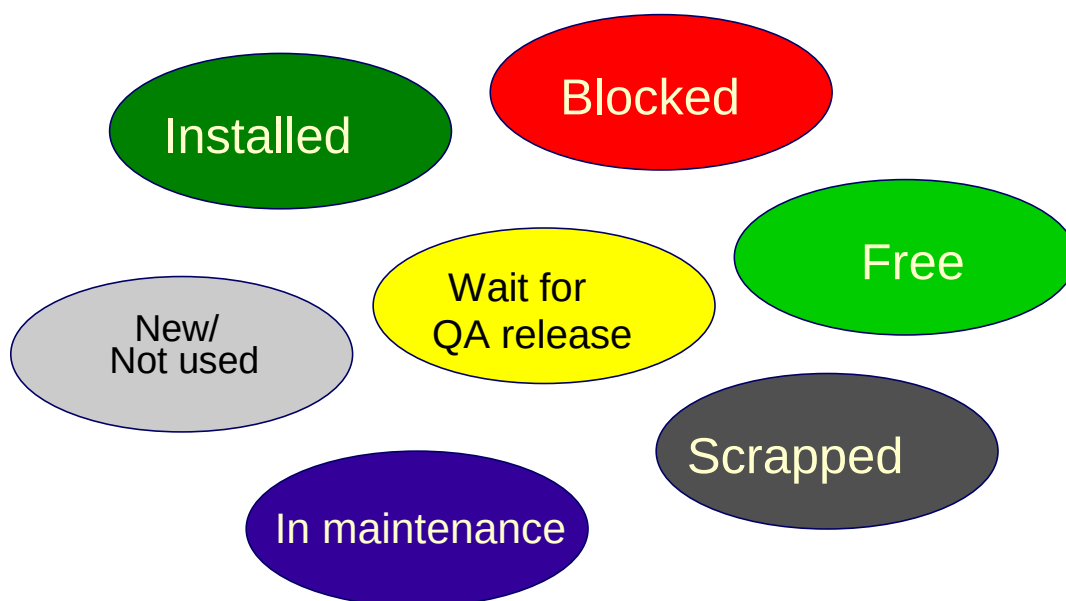
HYDRA menu	Master data → Resources → Resource status
FEDRA menu	Detailed Scheduling → Master data → Resource status
Transaction code	ressta
Function authorization	mdressta.*

This document describes the application "Resource status" on the client.

Purpose

Each resource has exactly one status at a time. You can configure any statuses for a resource. This way, it is possible to integrate a customer-specific status model.

Examples of status values:



You configure statuses for different resource types. You can define any status text you require. Different configuration attributes define the processing performed with the different statuses. For example, you can specify whether a resource can be logged on if a specific status is available.

In the *Status assignment*, you can optionally assign a storage location that is used in the *Set status* function. With status "Repair", you can then automatically assign the storage location "Tool shop", for example.

To set a status, use the function *Change status* in the applications *Resource overview* or *Resource stock*. You can assign a status immediately or for a future point in time. Setting or modifying a status is logged, so that the "resource history" shows the point in time when the status was changed.

Notes

With order logon, the resources are set to *Active* = Yes. The resource status does not change. The option *Collection* in the *Status assignment* configuration specifies whether a resource can be logged on or not. When the operation is logged off, the resource is reset to *Active* = No. Again, the resource status does not change.

The definition of a status transition model is not available in the status configuration. This means that the status can be changed as required, unless it is changed automatically by the system. At this point, we recommend to specify the relevant method.

The *active* flag is not relevant for resources of the type MNR. Even if an order is logged on to a machine, the resource status of the resource of type MNR remains *inactive*.

Optional assignment using resource families:

You can not only define a status using the resource type, you can also use resource families to assign a status. Configuration becomes more flexible this way. If you enter a value in field *Resource family*, this configuration is enabled. Important: All statuses must either be assigned using resource families OR resource types; otherwise this can lead to ambiguities!

Integration

You require the statuses to collect and document status changes. The defined status settings specify how a resource can be used. Example: If a resource is not allowed to log on, the operation in production cannot be logged on.

Selection criteria

In the selection panel, you can filter by higher-level or assigned resources. The application provides the following selection criteria:

Resource type

Type of resource

Resource family

Family the resource is assigned to.

Status

Status of the resource

Designation

Name of the resource status

Field descriptions

Resource type

Resource type for which the status is defined.

Resource family

Optional, if you want to define the status in relation to resource families.

Status

Status number; unique within the resource type or resource family.

Designation

Status name

Color

Color used to display the status in the *Resource overview*.

Note: The consistent use of the status/color combination with different resources is the responsibility of the customer.

Authorization

Authorization level for assigning a status on the terminal (a value between 0 and 9). In the HR master data, every person is assigned an authorization level for the resource status change. If the authorization level stored in the master data is lower than the authorization level defined here, you cannot assign the status via the terminal.

Note: Users who want to use the function *Change status* on the HYDRA client, require the relevant function authorization. With this function authorization, the status of a resource can always be changed; the authorization level on the client is *not* relevant.

Storage location

If a storage location is specified, then this value is automatically set in the function *Set status* (except if the storage location is explicitly set during status assignment, then this manually assigned storage location is used).

With status "Repair", you can then automatically assign the storage location "Tool shop", for example.

Assignment

This option controls whether the resource is displayed as locked or not locked in the *Graphic planning* (only if the additional function HLS-BSR is used).

Assignment = resource is displayed as locked

No assignment = resource is displayed as not locked

Collection

S = Resource is blocked for collection. In this case, a resource and also the operation cannot be logged on.

F = Resource is released for data collection and can be logged on.

Processing

Production identifier specifying the processing

F = Release:

This status is set with the automatic status monitoring, if the resource should be "released" again.

DNC:

When you create a new resource, the status with this identifier is only set as initial status, if no status with processing "I = initial status with creation" is configured for the resource type.

B = Status when logging OP off

This option is no longer supported since WRM 8.1.

O = Status when uploading optim. program (DNC)

This status is set for the upload of an optimized program.

U = Status when uploading (DNC)

This status is set for the upload of a program.

S = Automatic blocking status

If this status is set, the resource is locked.

L = Deleted (**from DNC 7.2**)

This status shows that the resource is no longer used, meaning it is deleted logically.

I = Initial status with creation (**from DNC 7.2**)

When creating a new resource, the status with this flag is assigned as an initial status. If this status is not available, then the released status is assigned to the resource.

Please note:

- For each resource type or resource family, you can only assign 1 status with one of these characteristics.
- When a resource is being directly logged off (no standard function), no status is set; in this case, the resource is only set from *active* to *inactive*.

Display of resource with this status on terminal

Display of the resources with this status on the Windows terminal.

Note: This configuration is only used in the DNC administration for resources of type DNC.

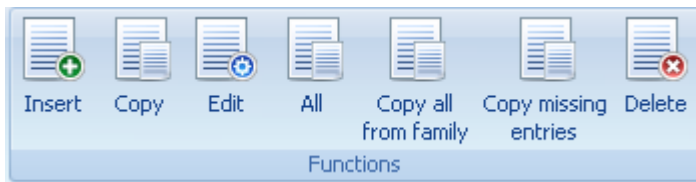
Deleted (from WRM 8.2)

If this option is checked, the resource has been (logically) deleted.

The identifier is used in the *Resource overview*.

Toolbar

The following functions can be called from the toolbar of the application.



Insert

Function authorization: mdrsstta.create

Opens the editing dialog to create a new entry.



Edit

Function authorization: mdrsstta.edit

Opens the editing dialog to change an existing entry.



Copy

Function authorization: mdrsstta.copy

Opens the editing dialog to copy an existing entry.



Delete

Function authorization: mdrsstta.delete

Opens the confirmation dialog to delete existing entries.



All

Function authorization: mdrsstta.copy

Opens the editing dialog to copy all entries of a resource type or a resource family.



Copy all from family

Function authorization: mdrsstta.copy

Opens the editing dialog to copy the statuses of a family.



Copy missing entries

Function authorization: mdrsstta.copy

Opens the editing dialog to copy the missing status assignments of a resource type.

7 Resources Posting Rules

Overview

Menu	Master data → Resources → Resources booking rules
Transaction code	resbr
Function authorization	resbr

This document provides a description of the "Resources booking rules" application in the Manufacturing Operation Center (MOC).

Usage

Flexible posting structures can be defined for energy management or for consumption entry. In addition to the fact that the BOM is displayed in a tree-like structure, offering selection options for report creation, there is also an option for compensation methods to be defined using booking paths. This option makes it possible to define multi-level images of calculated posted resources from counter resources that were entered either manually or automatically. In the process, the values can be added or subtracted as a percentage. Thus, in a sense a formula and an imaging path are defined. So, a source and a target resource as well as the compensation rule are always shown as a percentage. Random multiple relationships are possible, which allow the rules to be combined as formulas.

Integration

Compensation is a cyclical process, which creates documents based on the defined compensation rules based on documents that were already entered. Compensation is performed broken down at the most down to 10 steps. The target resource specified in a compensation rule includes the quantities and durations of the source resource added to or subtracted from the specified proportion.

Selection criteria

The application provides the following selection criteria:

Target resource type/ target resource

Definition of the target resource

Source resource type/ source resource

Definition of the source resource

Field descriptions

Target resource type/ target resource

Definition of the target resource

Source resource type/ source resource

Definition of the source resource

Valid from/ valid to

Rule's period of validity

Proportion

Proportion in percent, positive (added to) or negative (subtracted from) value

Toolbar**Paste**

Opens the dialog in which a compensation rule is entered.

**Copy**

Opens the dialog into which a compensation rule is copied.

**Edit**

Opens the dialog in which a compensation rule is edited.

**Delete**

Deletes one or several compensation rules.

8 Machine Counter Configuration

Overview

	→	
	→	
Menu	Master data → Workplaces/machines → Counter configuration	
Transaction code	ctr	
Function authorization	mdctr	

Purpose

The counter configuration defines the behavior of how the pulses transmitted by MSS (machine interface) should be interpret in terms of monitoring and quantity.

Integration

In addition to defining counter configuration, each time a machine is connected to the terminal several different settings must be configured.

Requirements

Before defining any configurations, you must first set up the [Machine](#).



Configuration changes

In order for the shop floor terminal program to be able to interpret the settings or changes that were made, you must first restart the terminal that is assigned to the workplace/ machine.

Selection criteria

The application provides the following selection criteria:

Machine

The ability to select the defined counter for the chosen resource/ machine.

Field descriptions

Machine

By entering the machine number, a logical counter channel is assigned to a machine.

Numerator

Counter number used to uniquely identify the counter channel for the machine.

Note that the assignment of the counter number to the physical connection to the CT-MSS, CT-UMPS or OPC variable depends on further settings (e.g. local terminal configuration file CTWIN.INI/CTAIP.INI (Windows terminal) or AIOP.CFG (DOS terminal)).

Designation (name)

Explanation of the counter, e.g. *Yield at closure counter*

Unit

Quantity unit that is recorded by the counter. Is currently only used for documentation purposes.

Cycle monitoring

If this identifier is set for a channel, incoming counter pulses are considered during cycle monitoring.

When calculating the actual cycle, only the first counter is referenced that is configured by the identifier "For monitoring". This counter must also be set for monitoring via operating signal if an actual cycle is to be calculated at the same time.

Posting as

Defines which quality is recorded at the counter quantity. Different values:

- yield
- scrap
- rework
- open quantity

The system posts generally to the primary quantity account.

Counters that are configured with the *No posting* option do not post anything to the quantity account. Only the counter reading is collected. Cycles or strokes may be posted and/or controlled via the cycle monitoring.

Reason

If you post quantities for scrap, rework or open quantities, you can store a reason in the counter. To do so, you must first create the reason in the configuration accordingly (*Reasons*).

Thus, for example, you can define two separate counters with different scrap reasons.



A scrap reason stored in the counterconfiguration has a higher priority than a scrap reason stored in the status.

Posting as cycles

If this option has been set, the pulses coming in via this counter are posted as cycles.

Posting with partitioning/ pulse factor

If this option is set, the incoming pulses are calculated using the respective active partitioning/ pulse factor.



For counters with the option *No posting*, the setting *Posting with partitioning/ pulse factor* is irrelevant, because no cycles are converted into quantities for these counters.

Offset against

You can offset the pulses recorded with this counter and the resulting quantities against a different quantity account. In this case, the entered quantity is deducted from the specified account.

Note: If you offset quantities, the resulting values can be negative values.



What you must be aware of for DOS based terminals is that you must make sure you do NOT set the option "Offset against" in the machine/ workplace configuration, index tab "Quantities", but instead you must set a compensation here.



Counters with the option "No posting" cannot be offset against quantity accounts.

Offset against material

This identifier is used to enter the counter that collects the material consumption of the input material. Which material is consumed from the BOM can be specified using either the material type or BOM item options.



Consumption recording

You can collect the material use in different ways. You can find a general overview here [Consumption recording](#).

Material type

If you enter a material type here, the counter is used for exactly this material of the BOM with this material type.

BOM items

If a BOM item is stored here, the counter is only used for this material.

Modified on

Modified by, date and modified on

Notes for DOS based terminals

The counter configuration is only supported to a limited extent for DOS based terminals (CT-56x, CT-541):



Only a maximum of one counter can be configured for each quantity account *Yield* or for each quantity account *Scrap* (no rework, no open quantity, and no counter with the option "No

posting").



An *Offset against* is only possible at the second counter that must be configured as *Scrap*. In this case, the compensation is only possible with *Yield*.

What you must be aware of is that you must make sure you do NOT set the option *Offset against* in the machine/ workplace configuration, index tab *Quantities*, if a compensation is set here.



No reason may be defined for the scrap counter.

Toolbar



Workplace configuration

Click this button to call the [Workplace and resource configuration](#).

9 Workplace terminal assignment

Overview

	→	
Menu	Master data => Workplaces/machines => Terminal assignment	
Transaction code	wta	
Function authorization	mdwta	

Purpose

This function is used to configure the machine assignment for the terminal.

Purpose

The assignment of a machine to a terminal is a requirement to use MDE-specific functions such as automatic shift change, cyclic status updates or recording via MSS. However, these functions are only available if the terminal is configured as a so-called "MDE terminal" (see Terminal Configuration). Terminals configured as ADE terminals do not provide these function even if the machines/workplaces are not assigned to the terminal.

The assignment ensures that the machine/OP is displayed by default on the terminal.

The number of machines that can be assigned depends on the terminal type.

Terminal type CT-541:	only one assignment possible
Terminal type CT-76x, CT 83x, CT84x, CT850 (AIP 8.1 and 8.2):	up to 16 assignments possible (even though the terminal is configured as master terminal) up to 10 machines may be used for the process data collection (HYDRA-PDV).
Terminal type CT-56x:	up to 8 assignments possible
Terminal type A-SUB	up to 20 assignments possible.

All machines previously assigned to terminals are shown according to the selected terminals. The order in which the display is shown on the terminal is determined by the position specified here.

If a production line (only available if MDE-LIN license is available) is assigned to the terminal, all aggregates of the production line are automatically assigned to the terminal and displayed in gray under the position "99". Aggregates cannot be removed from the assignment. If a production line is removed from the assignment, the aggregates assigned to the production line are automatically removed. Production lines can only be attributed to the terminal types CT76x, CT83x (max. 2 lines) and CT84x (max. 3 lines).

The option "Processing" can perform different assignments of the machine type to the terminal. The following options are available:

A - BDE processing

M – MDE processing

Processing as per operation mode of the terminal

Therefore, workplaces/machines with HYDRA-MDE-processing and workplaces only with HYDRA-ADE-processing may be assigned to an MDE terminal.

You have to set the processing to "BDE processing" for a group workplace, if you want to assign the group workplace to a terminal.



Number of terminal assignments

You can only assign a machine/workplace to a single terminal. If you want to assign a machine to several terminals, then you have to obtain the appropriate license.



Configuration changes

Restart the terminal to ensure that the settings or changes made can be interpreted by the terminal shop floor program.

Selection criteria

The application provides the following selection criteria:

From - to

Select terminal number

Field descriptions

Terminal

Assign a machine to a unique terminal number.

Position

Display position of the machine at the terminal and for the terminal assignment at the client.

Machine

Machine you assign to the terminal.

Processing

Here, you have the following 3 options:

- A BDE processing
- M MDE processing
- Processing as per operation mode of the terminal